Efficient Retrieval-Grounded Text Classification with Lightweight Transformers

Demo fixture — generated locally for PeerScope workflow demonstration only.

Summary

This paper studies efficient text classification with lightweight transformer models augmented by retrieval grounding. We describe a compact architecture, training procedure, and evaluation on two public benchmarks. Results show accuracy gains over a non-retrieval baseline with modest latency overhead. Limitations and future work are discussed.

1 Introduction

Text classification remains a core task in language processing. Large models are accurate but costly to deploy. This work investigates whether a small transformer combined with a retrieval component can close the gap. Our contribution is a lightweight architecture, a training recipe, and an empirical comparison against baselines.

2 Related Work

Prior work explores efficient transformers, distillation, and retrieval augmentation for knowledge tasks. We build on these directions and compare directly with strong baselines on shared benchmarks.

3 Method

Our model uses a 6-layer transformer encoder with hidden size 384. A retriever selects top-3 passages from a fixed index; embeddings are concatenated with the input representation before classification. Training uses cross-entropy loss, AdamW, learning rate 3e-5, batch size 64, 5 epochs, seeds 1-3. Data splits are fixed and documented.

4 Experiments

We evaluate on two public text-classification benchmarks with accuracy and macro-F1. Baselines include a linear model, a small transformer without retrieval, and a larger reference model. Each experiment is repeated with three seeds; we report means and standard deviations.

5 Results

The retrieval-grounded small model improves accuracy by 2.1 points and macro-F1 by 1.8 points over the non-retrieval baseline. Latency increases by 12% on CPU. Ablations show retrieval depth and encoder size both matter. Error analysis highlights long-tail classes.

6 Discussion

Gains are consistent but modest; retrieval quality bounds performance. Failure cases involve ambiguous labels and stale index entries. We discuss threats to validity including benchmark choice and hardware differences.

7 Conclusion

Lightweight transformers with retrieval grounding offer a practical accuracy-efficiency trade-off. Future work includes index refresh strategies and broader language coverage.

References

[1] Benchmark Author. Public text benchmark, 2023. [2] Efficient Models Author. Compact transformers, 2022. [3] Retrieval Methods Author. Retrieval for language tasks, 2023. [4] Evaluation Practice Author. Reliable evaluation, 2021. [5] Optimization Author. AdamW optimizer, 2019. [6] Statistics Author. Reporting with standard deviations, 2020. [7] Distillation Author. Knowledge distillation, 2021. [8] Indexing Author. Passage indexing, 2022.

Appendix: Reproducibility Notes

Code and data splits are described in Section 3-4. Hyperparameters, seeds, and compute environment (single CPU workstation) are documented. Results tables report means with standard deviations.